

Q4: Online Shopping Bill calculator

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Algorithm :-

1. Start
2. Input Quantity, Price, DisPercent, TaxPercent
3. IF Quantity < 1 , display "Invalid quantity" and stop.
4. IF Price ≤ 0 , display "Invalid price" and stop.
5. IF DisPercent < 0 OR DisPercent > 100 , display "Invalid discount" and stop.
6. IF TaxPercent < 0 , display "Invalid Tax" and stop.
7. Calculate SubTotal $\leftarrow$ Quantity $\times$ Price.
8. Calculate Discount $\leftarrow$ SubTotal $\times$ DisPercent / 100.
9. Calculate Amount $\leftarrow$ SubTotal - Discount.
10. Calculate Tax $\leftarrow$ Amount $\times$ TaxPercent / 100.
11. Calculate FinalBill $\leftarrow$ Amount + Tax.
12. Display Quantity, Price, DisPercent, TaxPercent, SubTotal, Discount, Amount, Tax, and FinalBill.
13. STOP.

Pseudo code

Begin

Let Quantity, Price, DisPercent, TaxPercent be user input.

IF (Quantity < 1), then

print "Invalid quantity"
STOP

ENDIF

IF (Price <= 0), then

print "Invalid price"
STOP

ENDIF

IF (DisPercent < 0 OR DisPercent > 100), then

print "Invalid discount"
STOP

ENDIF

IF (TaxPercent < 0), then

print "Invalid tax"
STOP

ENDIF

SUBTotal $\leftarrow$ Quantity * Price

Discount $\leftarrow$ SUBTotal * DisPercent / 100

Amount $\leftarrow$ SUBTotal - Discount

Tax $\leftarrow$ Amount * TaxPercent / 100

Final Bill $\leftarrow$ Amount + Tax

Display "Quantity: ", Quantity
Display "Price per item: ", Price
Display "Discount Percentage: ", DisPercent
Display "Tax Percentage: ", TaxPercent
Display "SubTotal: ", SubTotal
Display "Discount: ", Discount
Display "Amount after discount: ", Amount
Display "Tax: ", Tax
Display "Final Bill: ", FinalBill

STOP

